

ARUN SHARMA

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SUMMARY

I build physics-informed machine learning that generates, routes, and reasons about vehicle and mobility behavior reliably, including when signals are sparse, corrupted, or adversarially manipulated via diffusion models that generate physically realistic vehicle trajectories for traffic simulation and prediction, physics-informed detection of GPS-spoofed and deep-faked trajectories, and multi-objective routing over large road networks. My dissertation, Distortion-Aware Spatial Data Science, develops these methods with applications in autonomous vehicles and maritime security, urban mobility and travel behavior, climate, agriculture, and oncology, and through the USDA-NSF AI-CLIMATE and AI-LEAF institutes, I apply them to monitoring anomalous behavior in maritime and urban spaces and to estimating soil carbon, greenhouse gas, and crop yield with agronomy and soil science. Seeking full-time research opportunities.

EDUCATION

University of Minnesota, Twin Cities – Ph.D., Computer Science 2021 – 2026
Advisors: Prof. Shashi Shekhar | Committee: Prof. Vipin Kumar, Prof. Ravi Janardan, and Prof. Ying Song
Dissertation: Distortion-Aware Spatial Data Science | Doctoral Dissertation Fellowship (2022–23)

University of Minnesota, Twin Cities – M.S., Computer Science 2018 – 2021

TECHNICAL SKILLS

Languages: Python, Java, R, SQL, Scala, C/C++, Julia, Rust, Go, Shell (Bash), MATLAB, JavaScript (Node.js), TypeScript.

Machine Learning: MLflow, XGBoost, MLlib, LLMs (fine-tuning & inference), VLMs, RAG, LangChain, VectorDB (FAISS, Chroma, Pinecone), Hugging Face Transformers/TRL, LoRA, PEFT, RLHF, OpenAI API, Triton Inference Server, ONNX.

Agentic AI: Chain-of-Thought (CoT), ReAct, Tool/Function Calling, Agent Traceability, Tool-Call & Token Optimization, Multi-Agent Orchestration (LangGraph, AutoGen), MCP, Prompt Caching.

Big Data & Distributed Systems: PySpark, Flink, Presto, Airflow, Kafka, RabbitMQ, Delta Lake, Docker, Terraform, Kubernetes.

ML Frameworks & Deployments: PyTorch, TensorFlow, JAX, ONNX, SageMaker, Lambda, ECS, Bedrock, vLLM/SGLang, Temporal-Nexus.

RL & Alignment: RLHF/RLAIF/RLVR, PPO, DPO, GRPO, Constrained RL, Reward Hacking Mitigation, Agent Traceability, Tool-use Safety.

EXPERIENCE

ESRI – Environmental Systems Research Institute Research Scientist Intern, May 2023 – Dec 2023

- Improved detection of illegal vessel behavior (route deviations, dark shipping) from 55% to 73% accuracy by independently designing and deploying an end-to-end anomaly detection pipeline integrating Transformer-based models, Evidential Deep Learning (EDL), and AWS SageMaker on a large-scale maritime dataset (~500M AIS records), with automated retraining via Lambda, ECS, and Step Functions.
- Reduced maritime route query latency by 40% to support real-time vessel tracking by designing and implementing a scalable Graph-based Traffic Representation and Association (GTRA) framework leveraging PySpark and GeoAnalytics APIs over a large-scale AIS maritime dataset.

University of Minnesota, Twin Cities Graduate Research Assistant, Aug 2018 – PRESENT

- Led development of Pi-DPM, a physics-informed diffusion model that detects GPS-spoofed and AI deep-fake trajectories, pairing a transformer context-informed encoder with a physics-informed decoder embedding a single-axle Kinematic Bicycle Model prior and physics regularizers on velocity, acceleration, heading, curvature, and turning rate; outperformed six state-of-the-art baselines (DiffWave, DiffTraj, ControlTraj) at 5, 10, and 20 percent anomaly rates with within- and cross-domain transfer.
- Built GeoTrace-Agent, a production-grade multi-agent framework for auditable spatiotemporal reasoning over heterogeneous trajectory and Earth-observation sources, with a typed PlanGraph chain-of-thought planner that decomposes natural language spatial queries into deterministic sub-tasks over AIS, OSM road networks, Copernicus weather, Sentinel imagery, and space-time-prism tools, exposed through MCP and JSON-RPC A2A with OpenTelemetry tracing and human-in-the-loop review queues; instrumented per-query cost at about \$0.026 on a live evaluation run (Claude Sonnet 4.6), with an exact-response cache that serves repeated queries at zero model cost, and gated physically infeasible outputs.
- Designed Pi-GRPO, a physics-informed reinforcement-learning post-training stack that unifies PPO, DPO, and GRPO under a shared hybrid reward (a hard Kinematic Bicycle Model violation penalty, a jerk and curvature soft envelope, and a

Pi-DPM reconstruction-likelihood term); trained on about 11,000 preference triples, it reduced the hard-constraint violation rate from 18 percent under a preference-only reward to 0 percent under the physics-augmented reward while keeping KL divergence bounded over 3,000 PPO steps.

- Developed knowledge-guided machine learning for sustainable agriculture within the USDA-NSF AI-CLIMATE and AI-LEAF institutes, with Prof. David Mulla and Dr. Licheng Liu: SM-Hybrid, a hybrid spatial-neural-network surrogate for the DayCent agroecosystem model that captures within-site spatial autocorrelation and cross-site teleconnections and predicts land-surface carbon and greenhouse-gas fluxes (GPP, Ra, Rh, NEE) and crop yield across Iowa, Illinois, and Indiana about twice as fast as DayCent with higher accuracy than surrogate-ANN baselines; and a fine-tuning-based site-calibration method that transfers knowledge-guided models across agricultural sites.
- Generated realistic synthetic vehicle trajectories for autonomous driving simulation and traffic prediction by co-designing GCDM (Geo-lucid Conditional Diffusion Model), a two-stage cascaded generative diffusion injecting attributed road-network priors via map-informed latent variables, spatial cross-attention, and a physics-based trajectory decomposition (map-derived base + learned residual); produced trajectories with stronger geo-distribution similarity (density, trip error) and higher velocity/acceleration/jerk fidelity than 4 baselines (DiffTraj, Sashimi, DiffWave, EETG) on Porto (~1.7M trips) and Harbin taxi data, while reducing downstream traffic in/out-flow prediction error.
- Enabled real-time energy-vs-travel-time route planning for electric vehicles by co-designing MBOR (Multi-Level Bi-Objective Routing), introducing three novel data structures, a boundary multigraph, Multi-level Encoded Pareto Frontier View (MEPFV), and 2D cost-interval pruning with multi-edge pruning, to precompute and retrieve complete Pareto-optimal path sets over fragmented road networks; achieved >10x online-runtime improvement on real-world road network data for real-time bi-objective EV route planning.
- Reduced manual investigation area for illegal maritime activity (fishing, oil transfers, trans-shipments) by up to 80% by leading a multi-year, 4-publication research program developing STAGD (Space-Time Aware Gap Detection) with a Dynamic Region Merge (DRM) criterion combining comparison-less temporal indexing, R*-tree hierarchical spatial indexing, and maximal-union merge over space-time prism geo-ellipses; proposed TGARD and Dual Convergence DC-TGARD exploiting ellipse symmetry with bi-directional pruning and early-stopping; experimentally faster and more accurate than baselines on AIS data (500K ships) for homeland-security and public-safety applications.
- Knowledge-guided ML: Helped develop a fine-tuning-based site-calibration method that pretrains on the pooled regional signal and fine-tunes per field to estimate land emissions across Midwestern sites from remote-sensing primary production, climate, and soil covariates, framed as a spatial transfer-learning problem that adapts from sparse local data without losing interpretability.
- Spatial domain adaptation: Built methods that train where labeled data are dense and transfer where they are sparse, using spatially-aware representations, blocked spatial cross-validation, and calibrated uncertainty so output reflects out-of-region performance rather than spatial leakage.
- Physics-informed modeling and pipelines: Developed physics-informed generative and reinforcement-learning models that embed mass-balance and kinematic constraints into training (hard-constraint violations driven to zero), and built reproducible Python and R pipelines for large spatial and remote-sensing datasets on high-performance and distributed computing.
- Enabling precision immunotherapy through spatial analysis of tumor microenvironment: developed domain-adapted spatially-lucid neural networks for multi-category point set classification in non-Euclidean space, using a spatial ensemble framework with place-calibration parameters, weighted-distance learning rate, and spatial domain adaptation across heterogeneous tissue place-types. Leveraged a multi-task architecture with spatial mix-up masking and spatial contrastive predictive coding for self-supervised learning on MxIF oncology data to classify immune-tumor relationships for immunotherapy.
- Community-engaged spatial epidemiology for pandemic response: built an entity-relationship model, system architecture, and implementation to analyze aggregated mobile device data for understanding COVID-19 policy interventions on mobility. Delivered county-level analytics on long-duration visits to high-risk business categories and fine-resolution device count maps to epidemiologists, analysts, and policymakers. Collaborated directly with the Minnesota Department of Management and Budget; work cited in testimony to the MN House Transportation Finance Committee.

HONORS AND ACHIEVEMENTS

DOCTORAL DISSERTATION FELLOWSHIP

2022 – 2023

University of Minnesota, Twin Cities

TEACHING EXPERIENCE

SPATIAL DATA SCIENCE RESEARCH

Spring 2024

- Guest lectures on scientific methodology, fostering research skills, and effectively communicating research ideas.
- Supervised diverse projects and mentored multiple graduate and undergraduate students from interdisciplinary fields.

SPATIAL DATA SCIENCE

Fall 2019

- Responsible for handling class lectures, queries, homework assignments, labs, exams, lecture slides, etc.
- Guest lecture topics: Spatial Indexing, Networks, and Data Mining.

ADVANCED DATABASE SYSTEMS

Spring 2019

- Responsible for handling class lectures, queries, homework assignments, labs, exams, lecture slides, etc.
- Guest lecture topics: Concurrency Control, Database Security, and Data Mining.

DATA STRUCTURES AND ALGORITHMS

Fall 2018

- Instructed weekly recitation sessions with over 40+ students and graded 400+ students.

SERVICES AND LEADERSHIP

SUST 4096 SUSTAINABILITY INTERNSHIP, UNIVERSITY OF MINNESOTA, MN (Mentor)

Spring 2024

- Supervised an undergraduate student on a course project based on KGML for Sustainable and Precision Agriculture.
- Presented a lightning talk and a poster at the Annual AI-LEAF PI Meeting in UMN Saint Paul Campus, MN.

HONORS MENTORS CONNECTION, WAYZATA HIGH SCHOOL, MN (Mentor)

Fall 2023 – Spring 2024

- Advised a high school student to analyze a real-world case study of signal spoofing behavior in the maritime domain.
- The student helped analyze circular patterns, enabling us to create a taxonomy of spoofing behavior in open waters.

MINNESOTA DEPARTMENT OF MANAGEMENT AND BUDGET, MN (Service)

May 2020 – June 2021

- Reported county-level mobility traffic to epidemiologists, analysts, and policymakers for informed decision-making.
- Advised multiple high school and undergraduate students who were considering a research career.

INVITED PRESENTATIONS AND POSTERS

Poster: Towards Physics-guided Generative Foundation Models

2025

The 1st ACM SIGSPATIAL International Workshop on Generative and Agentic AI for Multi-Modality Space-Time.

Authors: Arun Sharma, Majid Farhadloo, Mingzhou Yang, Bharat Jayaprakash, William Northrop, and Shashi Shekhar

Poster: Spatial Distribution-Shift Aware Knowledge-Guided Machine Learning

2024

AAAI Bridge on Knowledge-Guided ML: Bridging Scientific Knowledge and AI, 2024.

Authors: Arun Sharma, Majid Farhadloo, Mingzhou Yang, Ruolei Zeng, Subhankar Ghosh, and Shashi Shekhar

Presentation: Towards Distortion-Aware Spatial Data Science

2023

Michigan Institute of Data Science and Society (MIDAS), University of Michigan, Ann Arbor

Authors: Arun Sharma and Shashi Shekhar

SELECTED PUBLICATIONS

[1] Towards Physics-informed Diffusion for Anomaly Detection in Trajectories: A Summary of Results. Proceedings of the 2nd ACM SIGSPATIAL International Workshop on Geospatial Anomaly Detection (GeoAnomalies '25). Arun Sharma, Mingzhou Yang, Majid Farhadloo, Subhankar Ghosh, Bharat Jayaprakash, and Shashi Shekhar.

[2] Towards Kriging-informed Conditional Diffusion for Regional Sea-Level Data Downscaling: A Summary of Results. 32nd International Conference on Advances in Geographic Information Systems (SIGSPATIAL '24). Subhankar Ghosh*, Arun Sharma*, Jayant Gupta, Aneesh Subramanian and Shashi Shekhar (*Both authors contributed equally to this paper).

[3] Physics-based Abnormal Trajectory-Gap Detection. ACM Transactions in Intelligent Systems and Technology, 2024. Arun Sharma, Subhankar Ghosh, and Shashi Shekhar.

[4] Analyzing Trajectory Gaps for Possible Rendezvous Regions. ACM Transactions in Intelligent Systems and Technology, 2022. Arun Sharma and Shashi Shekhar.

[5] Towards a Tighter Bound on Possible-Rendezvous Areas: Preliminary Results. 30th International Conference on Advances in Geographic Information Systems (SIGSPATIAL '22). Arun Sharma, Jayant Gupta, Subhankar Ghosh, and Shashi Shekhar.

[6] Geo-lucid Conditional Diffusion Models for High Physical Fidelity Trajectory Generation. Proceedings of the 33rd ACM International Conference on Advances in Geographic Information Systems (SIGSPATIAL '25). Mingzhou Yang, Arun Sharma, Majid Farhadloo, Bharat Jayaprakash, and Shashi Shekhar.

- [7] Towards Surrogate Models with Hybrid Spatial Neural Networks: A Summary of Results. Proceedings of the 8th ACM SIGSPATIAL International Workshop on Geospatial Simulation (GeoSIM '25), pp. 57–69, 2025. Shengya Zhang, Arun Sharma, Majid Farhadloo, Mingzhou Yang, Yao Zhang, Mu Hong, Licheng Liu, David Mulla, and Shashi Shekhar.
- [8] Towards Fine-Tuning-Based Site Calibration for Knowledge-Guided Machine Learning: A Summary of Results. 5th Annual AAAI Workshop on AI to Accelerate Science and Engineering (AI2ASE). Ruolei Zeng, Arun Sharma, Shuai An, Mingzhou Yang, Shengya Zhang, Licheng Liu, David Mulla, and Shashi Shekhar.
- [9] Towards Pareto-optimality with Multi-level Bi-objective Routing: A Summary of Results. Mingzhou Yang, Ruolei Zeng, Arun Sharma, Shunichi Sawamura, William F. Northrop, Shashi Shekhar. In Proceedings of 17th ACM SIGSPATIAL Intl. Workshop on Computational Transportation Science GenAI and Smart Mobility Session, pp. 36–45, 2024.
- [10] Spatially-Delineated Domain-Adapted AI Classification: An Application for Oncology Data. SIAM International Conference on Data Mining (SIAM SDM) 2025. Majid Farhadloo, Arun Sharma, Alexey Leontovich, Svetomir N. Markovic, and Shashi Shekhar.
- [11] Towards Spatially-Lucid AI Classification in Non-Euclidean Space: An Application for MxIF Oncology Data. SIAM International Conference on Data Mining (SIAM SDM) 2024. Majid Farhadloo, Arun Sharma, Jayant Gupta, Alexey Leontovich, Svetomir N. Markovic, and Shashi Shekhar.
- [12] Spatiotemporal Data Mining: A Survey. Handbook of Spatial Analysis for the Social Sciences, Edward Elgar, 2022. Arun Sharma, Zhe Jiang, and Shashi Shekhar.
- [13] Understanding COVID-19 effects on mobility: A community-engaged approach. 25th AGILE Conference on Geographic Information Science, 2022. Arun Sharma, Majid Farhadloo, Yan Li, Jayant Gupta, Aditya Kulkarni, and Shashi Shekhar.
- [14] WebGlobe: A cloud-based framework for interacting with climate data. International Workshop on Analytics for Big Geospatial Data (SIGSPATIAL) 2018. Arun Sharma, SM Arshad Zaidi, Varun Chandola, Melissa R Dumas, and Budhendra L Bhaduri.
- REVIEWER:** NeurIPS, ICML, ICLR, CVPR, ECCV, AAAI, IJCAI, SIGKDD, CIKM, ICDM, SDM, MLSys, SIGSPATIAL, TKDE, JMLR

REFERENCES (CONTACT INFORMATION)

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